Two-mode Network Analysis of Culture Concepts

Draft

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Abstract

Kroeber and Kluckhohn's classic 1952 monograph *Culture: A Critical Review of Concepts and Definitions* brought together over a hundred scholarly definitions of culture and indexed the key concepts they discuss. We build two networks from that index to explore how concepts and works are connected. In the *definitions network*, concepts like "society" and "beliefs" are linked to 167 definition codes, grouped into seven categories and credited to the author or authors that K&K index under each. In the *works network*, the same concepts are linked to the books and papers cited in those definitions. By comparing both networks, we can see which concepts span many scholars' work and which are concentrated in particular sources. We present basic statistics about the networks, visualize the data using correspondence analysis and spanning trees, and highlight the key concepts and works that anchor each network.

1 Data and networks

We build two networks from K&K's Index B. The *definitions network* links 470 concepts—concrete ideas like "society," "customs," and "beliefs"—to 167 definitions of culture, which K&K number as codes. Each code is credited to the author or authors K&K index under it, and falls into one of seven categories, which K&K label with the capital letters A through G: A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete. The *works network* links the same 470 concepts to 131 authored works (books, chapters, articles, and other publications), so we can see which ideas appear in which scholarly sources. Both networks use the same set of concepts, which lets us compare how the definitions and works overlap in what they discuss. Of the 167 codes, 165 appear in K&K's own author index. The other two, C-I-18a and D-II-16, are attributions cited in the main text of the book that the index does not list. We leave out K&K's Part III statement codes (C-5, D-5, D-7, E-17), which sit outside the Part II typology. Concepts have consistent names across the networks; where a concept appears under different headings in K&K's source text, we treat it as a single concept. Works are named by their author(s) and year, and a single work can be cited under multiple definitions.

The definitions network naturally breaks into 10 separate components, with the largest containing 622 nodes. The works network has 10 components, with the largest containing 589 nodes. Our analysis focuses on these largest components, which contain virtually all the meaningful connections. We use maximum spanning trees across multiple projections to identify the strongest connections in different views of the network.

2 Overview of the networks

Table 1 gives summary statistics for both networks. The first three rows show node and edge counts. The "degree" rows counts connections of each node type; a concept's degree is how many

definitions (or works) mention it, while a definition's or work's degree is how many concepts it contains. In both networks, concepts and definitions differ substantially in their average degree.

Table 1: Summary statistics for the two networks. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors.

	Definitions network	Works network
Concept nodes	470	470
Definitions / works	167 codes (5 isolated)	131 works (4 isolated)
Edges	1189	1147
Density	0.015	0.019
Mean degree, concepts	2.53 (range 1–36)	2.44 (range 0–33)
Mean degree, definitions/works	7.12 (range 0–25)	8.76 (range 0–51)
Mean k_{nn}, concepts	10.32	14.46
Mean k_{nn}, definitions/works	9.01	8.63
Degree correlation r	−0.074	−0.163

We see a notable pattern in the lower rows of the table documenting the degree correlation in each network; concepts with many connections tend to link to definitions or works with few, and vice versa. The nearest-neighbor rows show the other side of this pattern. A concept's neighbors average 10.3 (definitions network) or 14.5 (works network) uses, while a definition's or work's neighbors average only about 9 concepts, so concepts attach preferentially to heavily used nodes. The works network is also slightly denser (0.019 against 0.015) despite a similar edge count (1,147 against 1,189), because it has fewer second-mode nodes. Finally, the maximum work degree of 51 belongs to Kelly & Kluckhohn 1945a, a fact that returns in the spanning-tree analysis below.

Figure 1 shows the degree distribution plots for each set of nodes in each network. Most concepts are lightly connected: about 79% appear in two or fewer definitions, and about 80% are cited by two or fewer works. In contrast, definitions and works are more heavily connected, with the average definition covering 7 concepts (median 7) and the average work covering 9 concepts (median 8), and some definitions and works discussing many more. This asymmetry means that a few concepts are discussed widely, while most concepts appear in just one or two places. The two second-mode distributions differ in their tails: works reach degree 51 while definitions top out at 25, so a single work can cover far more concepts than any single definition. The concept distributions, by contrast, are nearly identical across the two networks, as expected given that both share the same 470 concepts.

3 Maximum spanning trees

A maximum spanning tree (MST) keeps the strongest ties in a network while still connecting every node, using as few links as possible. A two-mode network has no ties between two concepts or between two definitions, so we analyze it through multiple one-mode projections [Everett and Borgatti, 2013]. Following Breiger [1974], two concepts are tied if they share at least one definition or work, with weight equal to the number they share. Likewise, two definitions (or works) are tied through the concepts they share. In matrix form, the projections are:

$$W_{\text{concept}} = AA^{\top}, \qquad W_{\text{def}} = A^{\top}A,$$

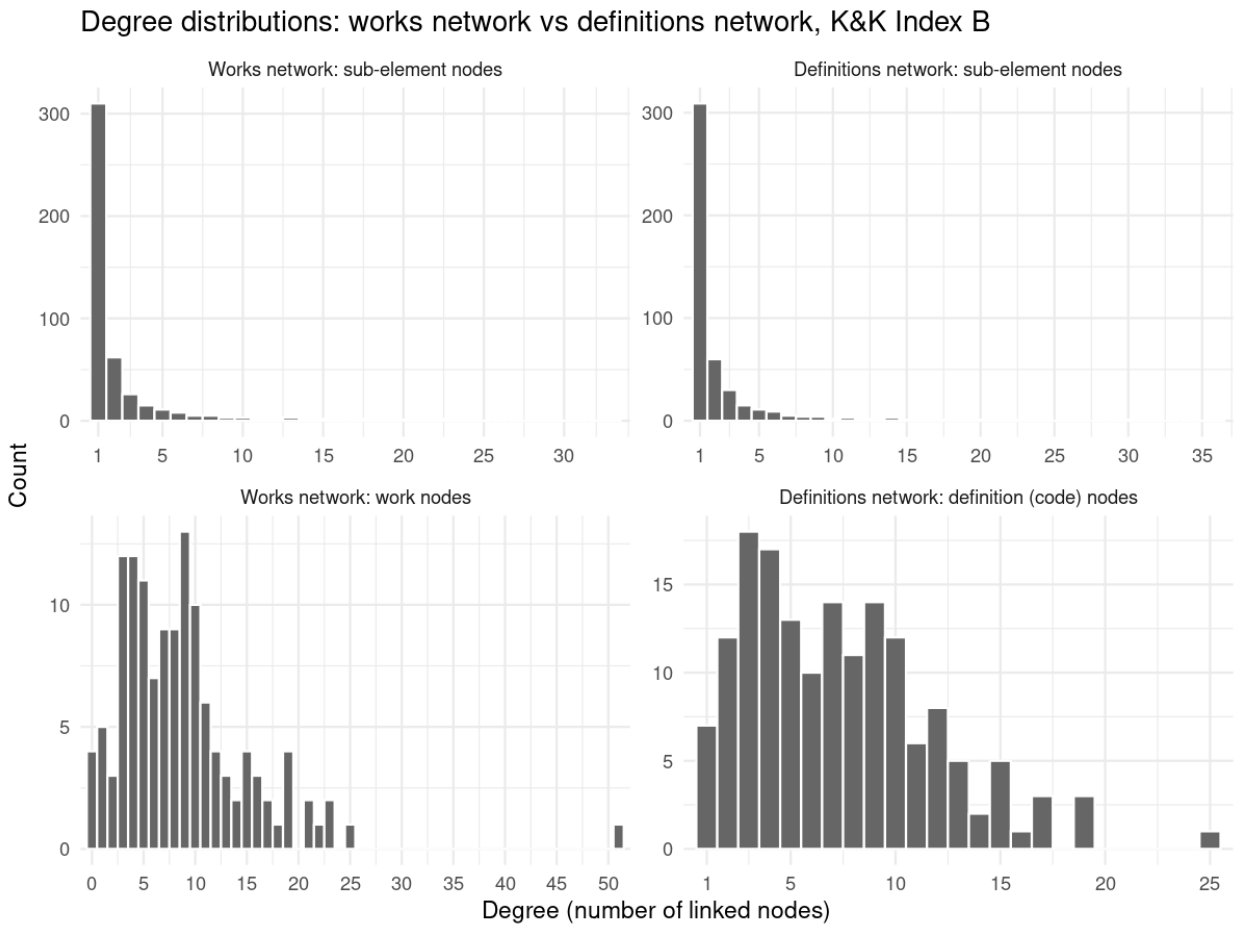


Figure 1: Degree distributions. Top row: sub-elements in the works network (left) and the definitions network (right). Bottom row: works (left) and definition codes (right).

where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ counts the definitions that include both concepts i and j. Each projection is a weighted graph whose edges are the off-diagonal pairs with positive weight. We build projections from the 2-core of each network's largest component.

The maximum spanning tree of each projection is found by computing the minimum spanning tree on negated weights, $-w$. Kruskal [1956] sorts edges from largest weight to smallest and adds each unless it would close a cycle, producing a tree that maximizes $\sum_{e \in T} w_e$. Because the 2-core is connected, each tree has exactly $|V| - 1$ edges. In the visualizations, node size shows betweenness within the tree and node color shows the community structure detected using Newman's leading-eigenvector method applied to the tree.

The definitions network has two main projections. The concept-level tree (Figure 2) shows how 159 concepts cluster. The definition-level tree (Figure 3) shows how 145 definitions cluster. Table 2 lists the top 10 concepts and definitions by betweenness in their maximum spanning trees.

In the concept tree (Figure 2), the four largest nodes—social, group, customs, and behavior—are the top four of Table 2. Social heads a green region of social-organizational vocabulary (social heritage, institutions, customs, social group), society, fifth in the table, anchors a pink region of learned and shared behavior (learned behavior, habits, socially transmitted), and behavior and group bridge the two. Beliefs, sixth in the table, heads the teal region of norms, knowledge, and

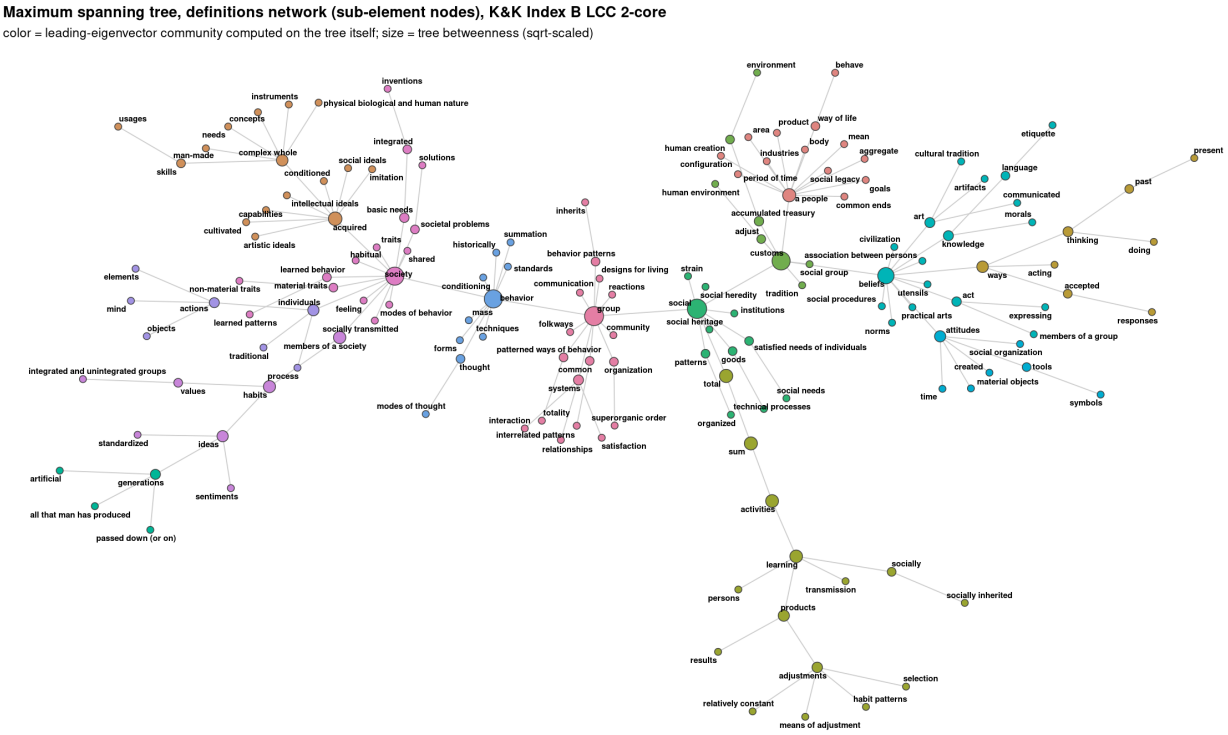


Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

artifacts. The tree also has one long peripheral branch running from total and sum down through activities, learning, transmission, products, and adjustments; this chain collects the acquisition vocabulary of the Genetic definitions and hangs off the core through a single path, so these ideas connect to the rest of the network only transitively.

The definition tree (Figure 3) puts its three largest nodes, F-I-3, C-I-6, and F-I-8, at the center of the tree. C-I-6 heads the largest community (23 definitions drawn from six of the seven groups), while F-I-3 and F-I-8 sit in the neighboring community, which is mostly Genetic. This is consistent with the table, where eight of the top ten definitions carry a C or F code. Degree and betweenness dissociate in this ranking: C-I-6 combines the second-highest betweenness with the highest tree degree (12), acting as the main hub of that community, while F-I-3 reaches the highest betweenness with only four ties, sitting on the paths that join the community to the rest of the tree. One author supplies a striking share of the bridges, as Bidney holds three of the ten most central definitions (C-II-3, C-II-4, and A-16). Away from the center, the Descriptive (A) definitions form a separate orange community in the upper left around A-3, and a chain of Descriptive and Historical definitions (A-16, C-I-4, A-1, A-14) runs along the right side, showing that the earliest and simplest definitions connect to the core through fewer, longer paths.

The works network also has two main projections. The concept-level tree (Figure 4) shows how 156 concepts cluster within the works network. The work-level tree (Figure 5) shows how 118 works connect through shared concepts. Table 3 lists the top 10 concepts and works by betweenness centrality.

The works-network concept tree (Figure 4) resembles its definitions-network counterpart but with a sharper hierarchy. Group is the largest node, at the center of a large green region of community, body, and ways vocabulary. Social, a pink region of social organization (patterns,

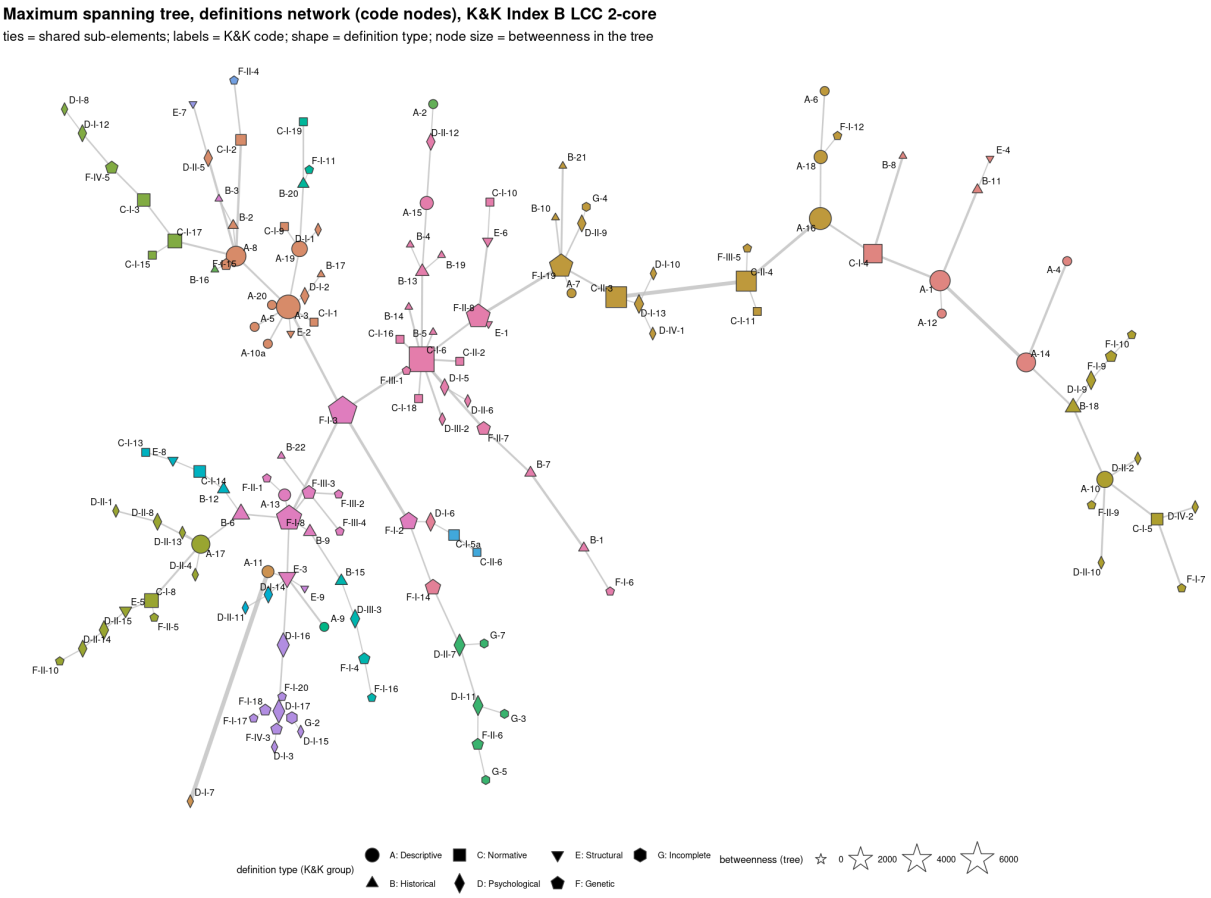


Figure 3: Maximum spanning tree of the definitions network, code (definition) nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

institutions, beliefs, customs), sits just beside it and has the second-highest betweenness; its tree degree (18) is second only to society (21), which heads an orange region at the upper left together with way of life, a people, and members of a society. Ideas heads a lime region at the right, with learning, habits, generations, and passed down, and total sits at the bottom with sum and civilization. The acquisition vocabulary of the Genetic definitions does not form a separate chain here: it appears as a short branch at the upper left (acquired, learned patterns, process).

The work tree (Figure 5) has a single dominant center. Kelly & Kluckhohn 1945a, the largest node and top entry of Table 3, sits at the middle of the plot with eighteen spokes reaching into most color regions, so many paths between works on opposite sides of the tree pass through it; it is at once the largest hub and the strongest bridge. Gillin 1942, Young 1934, Winston 1933, and Folsom 1931, the next four works in the table, appear as secondary hubs, each heading a community of its own, and Linton 1936 is the only Linton in the top ten. Most works, by contrast, are small peripheral leaves attached by a single edge, visible as the many small nodes at the plot's edges; Willey 1929, for example, despite its central position in the drawing, is one such leaf, a reminder that the spring layout places nodes by topology rather than importance.

Note to Table 2: Definition codes use the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it.

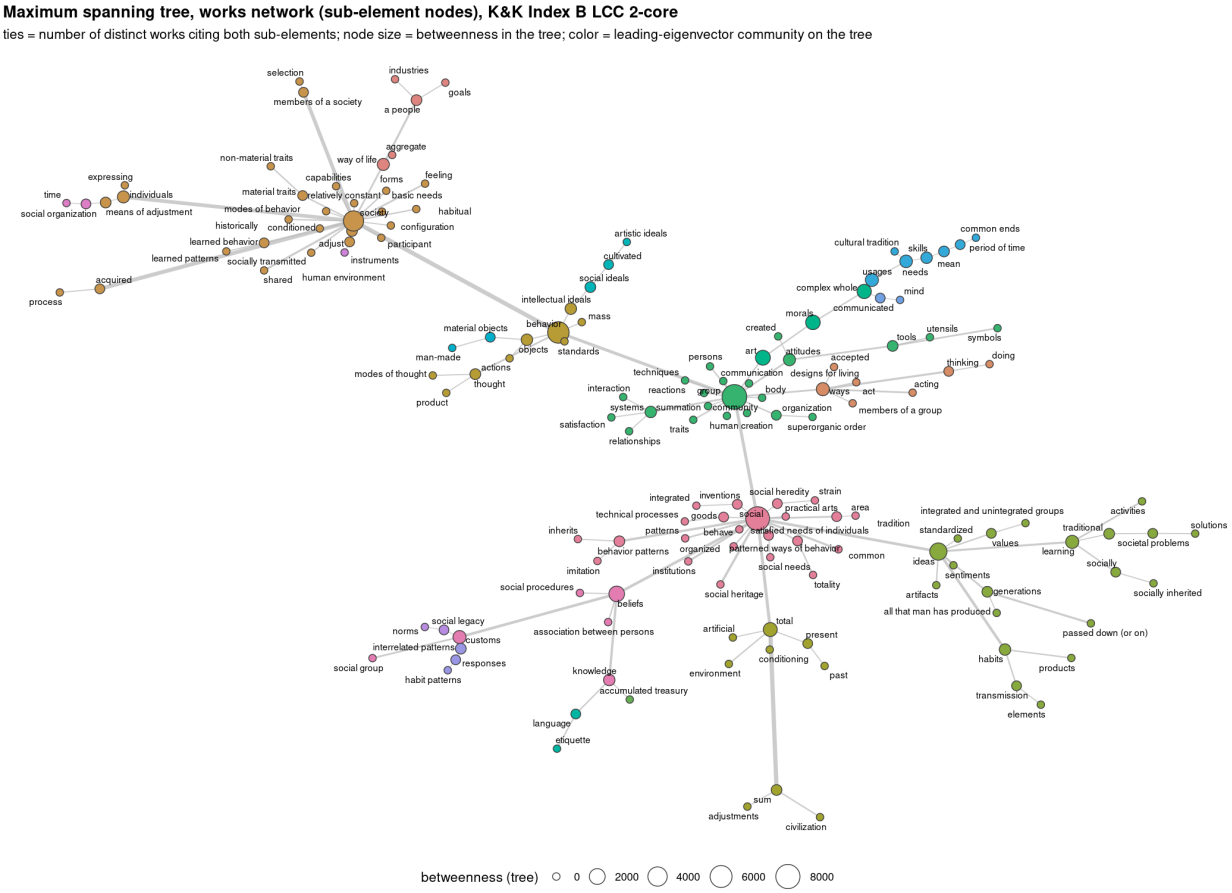


Figure 4: Maximum spanning tree of the works network, sub-element (concept) nodes. Ties count the works that cite both sub-elements.

Taken together, the four trees tell a consistent story. Six of the ten most central concepts are the same in both networks—social, group, society, behavior, beliefs, and total—and each of the leading concepts anchors one of the color regions of the concept trees, so the conceptual core of the culture concept is stable whether it is measured on definitions or on the works that cite them (customs, acquired, a people, and sum enter the definitions ranking; ideas, art, morals, and complex whole enter the works ranking). The two networks differ in how concentrated their second mode is. Among definitions, centrality is spread fairly evenly: the top definition (F-I-3, 69%) and the top concept (social, 63%) reach similar normalized betweenness, and no single definition approaches gatekeeper status. Among works, centrality concentrates at a single point, Kelly & Kluckhohn 1945a at 75% of the maximum possible, with the runner-up at 61%; the works literature is more star-shaped around one synthetic review than the definitions are around any of their members. In both concept trees the acquisition and transmission vocabulary sits on peripheral branches rather than at the core, suggesting that Genetic concerns, however common in the definitions themselves, connect to the rest of the conceptual structure only through a narrow path.

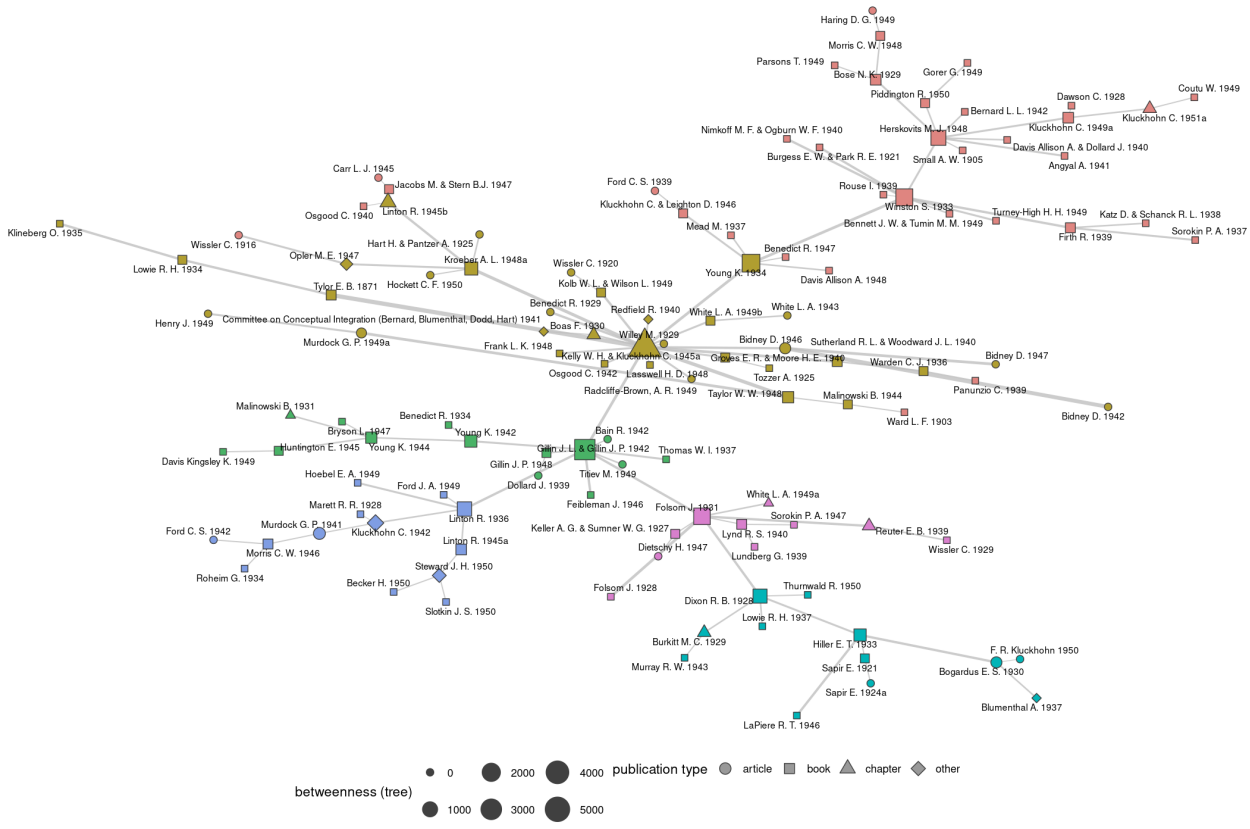


Figure 5: Maximum spanning tree of the works network, work nodes. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

4 Data-driven types: non-negative matrix factorization

The spanning trees and the community colors above classify nodes hard: each work belongs to exactly one community of the tree. Non-negative matrix factorization (NMF) [Lee and Seung, 2001] offers a soft, parts-based alternative. It approximates a non-negative matrix V by a product WH with all entries non-negative, so every node carries a loading on each factor and every factor is an additive mixture of the original features, with no subtractive terms. We run two NMF analyses on the works network, one on the coding structure and one on the wording of the definition statements, and ask in each case how far the data-driven types agree with K&K's own seven categories A–G. All use the same base-R implementation (Euclidean multiplicative updates, deterministic given a seed, with ten random restarts per value of k), and all fix k at seven, one factor per K&K group. Fixing k this way is a choice, not a result: the reconstruction error falls monotonically in k without an elbow, and restart stability is flat beyond $k = 2$, so neither criterion selects a number of factors on its own.

4.1 Types from the coding structure

The first analysis factorizes the 156×118 sub-elements $\times$ works matrix of the works 2-core, the same matrix the work tree of Figure 5 was built from. Table 4 gives the top sub-elements of

Table 2: Top 10 nodes by Freeman-normalized betweenness in the definitions network MST.

Node type	Node	Norm. Betw. (%)	Degree
Concepts	social	62.94	10
	group	59.53	12
	customs	51.79	7
	behavior	51.00	10
	society	50.92	14
	beliefs	35.99	12
	acquired	17.92	9
	total	16.25	2
	a people	15.82	13
	sum	15.20	2
Definitions	F-I-3	69.14	4
	C-I-6	57.54	12
	F-I-8	48.46	6
	F-II-8	40.84	4
	F-I-19	38.60	6
	A-3	34.04	9
	C-II-3	33.15	3
	C-II-4	29.35	4
	A-16	26.62	3
	C-I-4	22.19	3

each of the seven factors, which we label as follows: a *group behavior* factor (group, behavior, ways, common, material traits), a *norms and knowledge* factor (beliefs, knowledge, customs, art, complex whole, language), a *way of life* factor (a people, way of life, environment, total, goals), a *society–individual behavior* factor (society, acquired, behavior, individuals, members of a society), a *learning and acquisition* factor (ideas, habits, learning, transmission, generations), an enumerative *complex whole* factor (sum, total, beliefs, tools, activities), and a *social organization* factor (social, customs, institutions, social heritage, patterns). Figure 6 shows each work's loadings, rescaled to sum to one per work, with works grouped by their dominant factor. The dominant-factor split of the 118 works is 17, 20, 10, 21, 21, 14, and 15. Most works draw on more than one factor: at the 0.10 threshold used below, only 16 of the 118 works fall in a single factor.

To compare these types with K&K's categories, each work inherits the group of its codes (the letter A–G of the codes it is cited under; a work's dominant group is its plurality letter). This label is well defined: 112 of the 131 works are cited only under codes from one group, so the comparison is not an artifact of works straddling categories. The agreement between the dominant factor and the dominant group is at the level of chance: adjusted Rand index 0.016 against a permutation baseline whose 95th percentile is 0.026, and no factor maps onto any group. The largest cell in the whole cross-tabulation is six works.

4.2 A spanning tree of the factor types

The works factors can be hardened into a network the way the trees above were built. Binarizing the loadings at 0.10 turns each work into a member of the factors it loads on substantially: 16 of the 118 works belong to a single factor, 51 to two, 41 to three, 9 to four, and one to five, and no work drops out. Projecting the resulting works $\times$ factors biadjacency ties two works by the number of factors they share, giving 4,383 ties in a single component, and the maximum spanning tree of

Table 3: Top 10 nodes by Freeman-normalized betweenness in the works network MST.

Node type	Node	Norm. Betw. (%)	Degree
Concepts	group	71.30	17
	social	63.71	18
	behavior	47.37	7
	society	38.94	21
	ideas	22.82	8
	beliefs	15.89	5
	art	13.27	2
	morals	12.15	2
	complex whole	11.13	3
	total	10.05	6
Works	Kelly & Kluckhohn 1945a	75.39	18
	Gillin 1942	60.55	9
	Young 1934	39.51	6
	Winston 1933	34.00	8
	Folsom 1931	31.76	7
	Herskovits 1948	22.40	8
	Linton 1936	19.23	5
	Dixon 1928	17.67	5
	Kroeber 1948a	13.16	5
	Hiller 1933	9.98	4

that projection (Figure 7) again has $|V| - 1$ edges. The threshold, like k, is a choice rather than a result: lower it and the projection densifies; raise it to 0.3 and 71 of the 118 works fall back to a single factor, while one drops out entirely.

The tree's bridges are largely not the concept tree's. Kelly & Kluckhohn 1945a, the dominant center of the concept-based work tree (Table 3), is a leaf here, tied to a single neighbor. In the other direction, Parsons 1949, the strongest bridge of the factor tree, is a leaf of the concept-based tree, and Osgood 1942 and Slotkin 1950, also central here, are leaves there. Winston 1933 and Linton 1936 are central in both trees, the only works in both top-ten lists. The two community partitions agree only weakly (adjusted Rand index 0.06), seventeen communities covering this tree against six in the concept-based one. Neither tree is wrong: the two tie definitions ask different questions. A work is central in the concept tree when it draws on many distinct sub-elements that other works also draw on; it is central in the factor tree when its pattern of concept use sits between the factor profiles of other works. The inversion is itself evidence for the reading below—the position of a work in the corpus depends on what is counted as a tie, and no tie definition so far recovers K&K's groups.

4.3 Types from the wording of the definitions

The second analysis asks the same question of the definition texts themselves, independently of K&K's coding. The corpus holds 167 definition codes carrying a quotation: the Table 2 texts mapped to K&K codes through the crosswalk, the appendix-only texts, and one statement K&K prints itself (a Part II page footnote). Documents average about thirty words, so the term–document matrix is built restrictively: tokens of three letters or more, a stopword list that removes the words nearly every definition contains (culture, society, social, man, and the general definitional vocabulary total, whole, learned, acquired), and terms kept to a document frequency between

Table 4: Top five sub-elements of each factor, works NMF at $k = 7$ (loadings normalized to unit sum).

Factor	1st	2nd	3rd	4th	5th
Group behavior	group	behavior	ways	common	material traits
Norms and knowledge	beliefs	knowledge	customs	art	complex whole
Way of life	a people	way of life	environment	total	goals
Society–individual	society	acquired	behavior	individuals	members of a society
Learning and acquisition	ideas	habits	learning	transmission	generations
Complex whole	sum	total	beliefs	tools	activities
Social organization	social	customs	institutions	social heritage	patterns

two documents and half the corpus, leaving 402 terms weighted by TF–IDF. The four quotations K&K and Boroch print in French or German enter as English translations (K&K's own rendering for Maquet, ours for the other three, verified against the printed originals). As with the coding structure, the factorization is run at $k = 7$, one factor per K&K group.

The factors again find recognizable themes rather than K&K's categories (Figure 8), and we label them by their leading terms: *symbolic systems* (concrete, process, symbolic, systems, interaction), *material vs. non-material* (material, non, traits, associated, possesses), *species and utensils* (events, species, order, dependent, utensils), *goals and persons* (certain, rather, others, persons, goals), *feelings and adjustment* (inevitable, feeling, thinking, equipment, external), *passed down generations* (generation, common, sum, call, down), and *mental and holistic* (concept, mean, holistic, traits, mental). On a corpus this small a factor can be driven by one author's phrasing—the literal word "Kluckhohn" is among the top terms of the feelings-and-adjustment factor—so the labels should be read as descriptive of the top terms, not as clean concepts. Against the A–G groups the agreement is again chance-level (adjusted Rand index 0.018, permutation baseline 95th percentile 0.016), with no factor exceeding 54% purity in any group.

4.4 Interpretation

Neither signal recovers K&K's taxonomy: not the works' pattern of concept use, not the wording of the definition statements. Two readings are possible. One is that K&K's groups classify the emphases of individual *statements*, and that these emphases are lost when statements are aggregated to the works that contain them; against this stands the fact that works overwhelmingly sit within a single group, so the aggregation mixes little. The other, which we find more plausible, is that the seven groups are editorial categories—useful for organizing a bibliography—rather than dimensions along which the corpus varies. This is consistent with the spanning-tree picture above, where the vocabulary the Genetic group concerns itself with sits on a peripheral chain of the concept trees rather than forming a region of its own, and with the community structure of the trees, which follows concept vocabulary, not authorial schools. Hardening the works factors into a tree (Figure 7) does not change the verdict: its bridges are largely a different set of works, its communities do not follow A–G either, and the disagreement between the two tie definitions reinforces the point that no single structural reading of the corpus reproduces the taxonomy. The null result should be read against the method's limits as well: the number of factors was fixed to match the taxonomy under test rather than selected by the data, the short corpus makes TF–IDF factors fragile (restart agreement ranges from 0.44 to 0.76 for k from 3 to 8), and the texts are K&K's own selections rather than

the works' full statements.

5 Limitations

The networks depend on the K&K Index B transcription and on the verified-code filter. Element names are not unique across headings, so same-named elements are merged, which can join unrelated entries. Works are keyed by author and year, so a work with several year forms may appear twice. Several readings depend on OCR corrections recorded in the repository's manual-fix files; the validators check structure and round-trips, not the content of each edge. Two codes that are not in Index A are included: C-I-18a and D-II-16, from the main text. K&K's Part III statements (C-5, D-5, D-7, E-17) are excluded. The basis for each is documented in the repository.

A Supplementary sankey diagrams at the subgroup level

The NMF comparisons in the body classify against K&K's seven group letters. The two figures here repeat the comparison at the finer grain of K&K's own typology, the roman-numeral subgroups the classification prints within groups C, D, and F; groups A, B, E, and G number their codes plainly and have no subgroup level. A work's subgroup is the plurality of the subgroups of the codes it is cited under, and a work whose codes tie across subgroups is collected in the grey node—16 of the 118 works of the 2-core. Each ribbon takes the color of its origin node's group, and node size is the number of works or codes.

On the left, B reaches all seven factors, and A, C-I, D-I, D-II, and F-II each reach six; the grey node also reaches six, though it is a catch-all by construction. On the right, *group behavior* draws most on the grey node (4 of its 17 works), with B, C-I, and D-I contributing 2 each. *Norms and knowledge* is supplied mainly by A and C-I, with 5 works each. *Way of life* draws 3 works from the grey node and 2 each from B and D-I. *Learning and acquisition* draws 4 works each from B and D-II. *Society-individual behavior*, with 21 works, draws 3 each from C-I and the grey node and 2 from B, with no subgroup supplying more than three. *Complex whole* draws mostly on F-I (5 of its 14 works) and A (3), and *social organization* on B and E, with 3 works each. Cells of one to three works are too small to support more than a tendency.

At the subgroup grain the scatter does not tighten: the largest cells hold five works (A and C-I on *norms and knowledge*, and F-I on *complex whole*), against six at the group level. The larger subgroups reach five to seven of the seven factors, while E, F-III, and G, with five or six works each, reach three or four, and the small subgroups C-II, D-III, and D-IV reach one to three.

On the left, A places codes on all seven factors, and B, D-II, and F-I each reach six. On the right, *symbolic systems* draws mostly on F-I (10 of its 21 codes), with B (6) and D-I (5). *Passed down generations*, the largest factor at 48 codes, draws on A and C-I (8 each), B (6), and D-I and F-I (5 each). *Feelings and adjustment* is mostly D-I (6 of its 17 codes, 46% of the factor). *Species and utensils* draws on C-I (5 codes) and B (4), and *goals and persons* on D-II (4 of its 16 codes). *Material vs. non-material* draws on A and G (4 codes each) and D-II (3).

For the definition codes no aggregation is needed, since each code carries its own subgroup. The finer grain shows somewhat more concentration than the works panel: the largest cell holds ten codes (F-I on *symbolic systems*, 10 of 21), and A and C-I each place eight codes on *passed down generations*. No subgroup's codes sit on a single factor, and the factor that collects the most codes over all, *passed down generations*, draws from every group letter.

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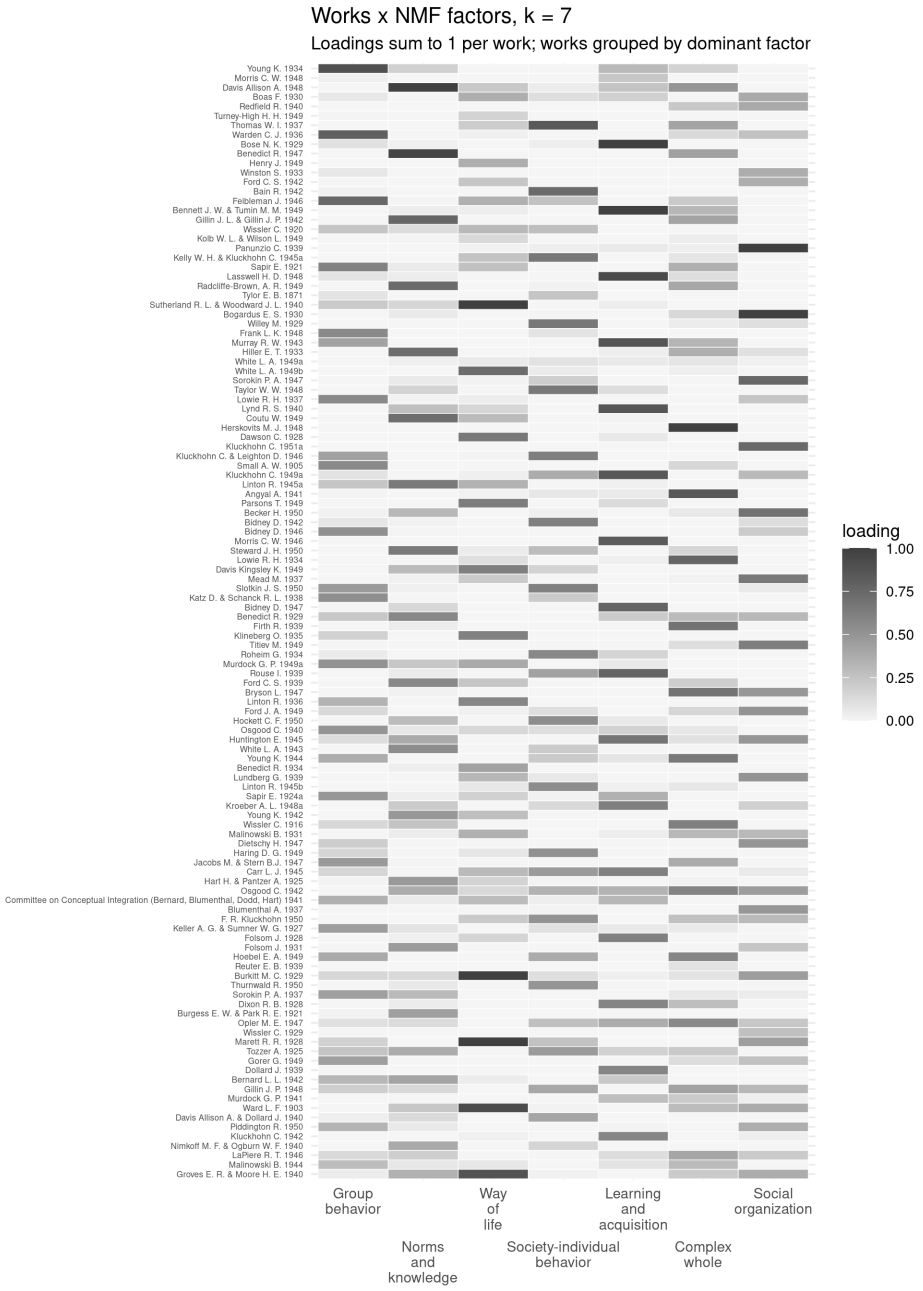


Figure 6: Works × NMF factors at $k = 7$. Each work's loadings sum to one; works are grouped by their dominant factor, ordered within each group by that loading. Factors are labeled by their leading sub-elements (Table 4).

Maximum spanning tree, works network (work nodes), NMF factors

binarized works x factor loadings (threshold 0.10); ties = number of NMF factors shared; node size = betweenness; color = community on the tree; shape = publication type

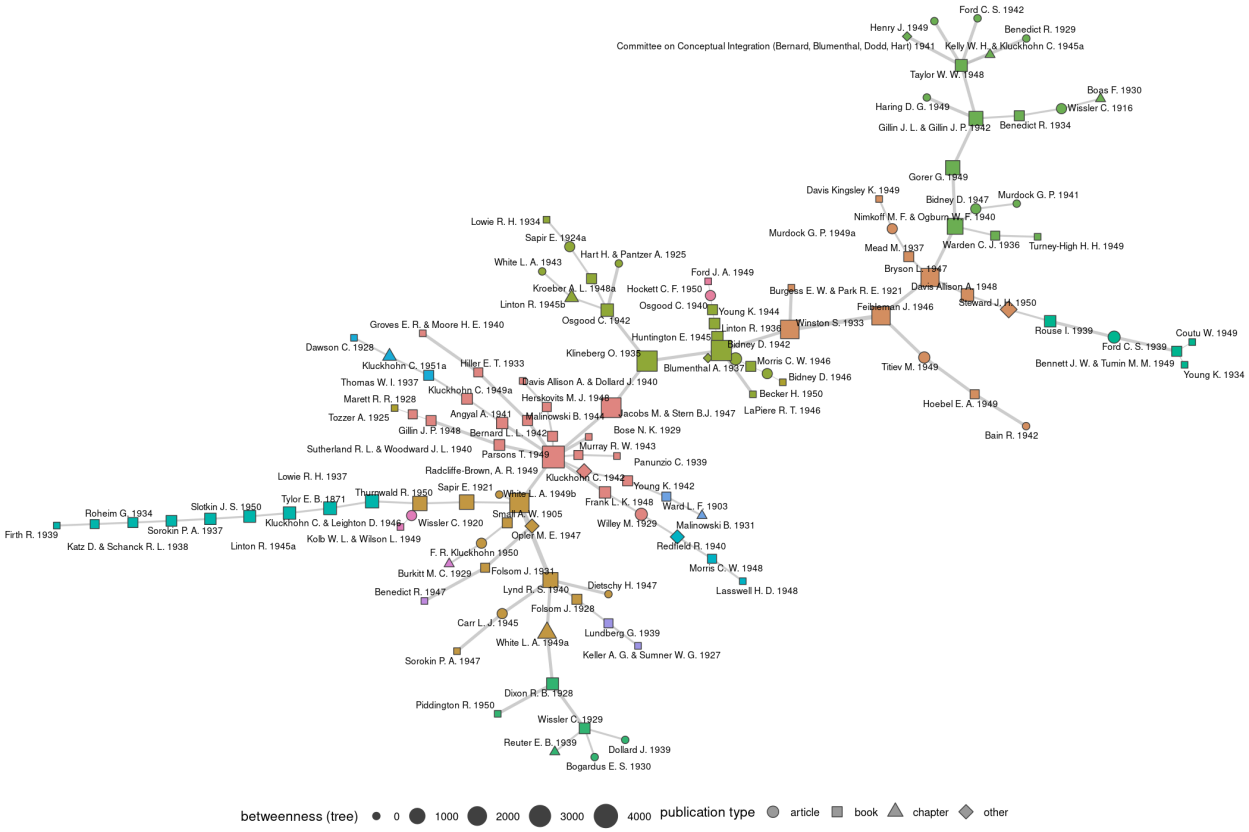


Figure 7: Maximum spanning tree of the works × NMF-factor projection (threshold 0.10), work nodes. Ties count the factors two works share. Color is the leading-eigenvector community on the tree; size is betweenness in the tree.

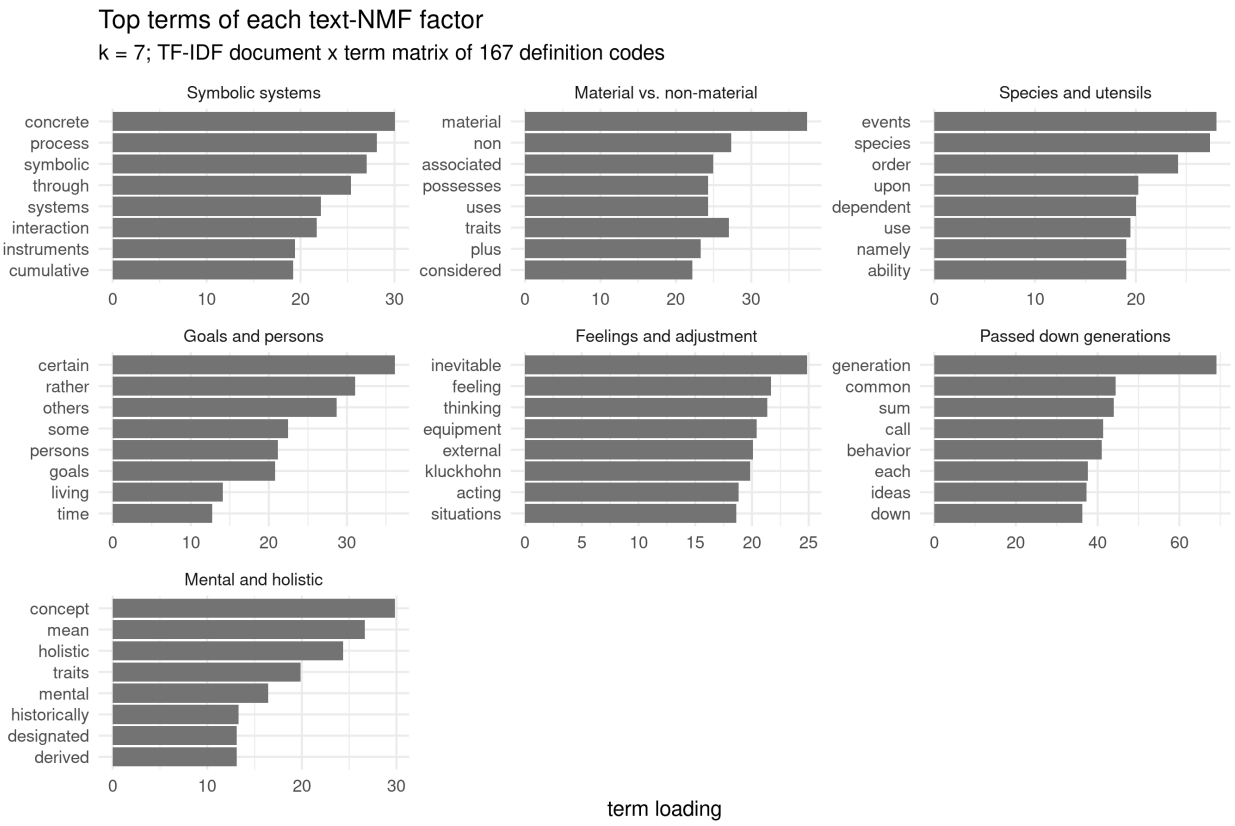


Figure 8: Top terms of each factor of the text NMF at $k = 7$, from the TF–IDF matrix of the 167 definition codes.

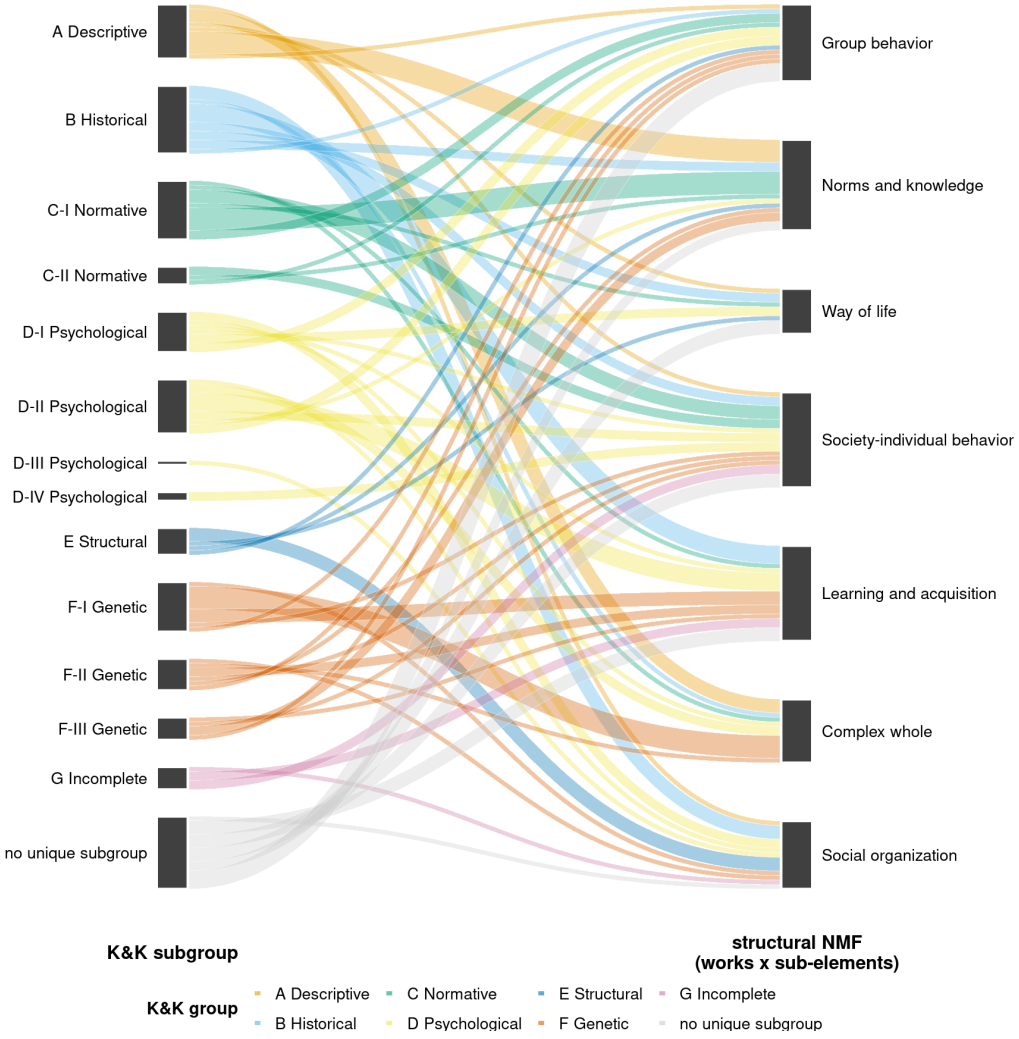


Figure 9: Works of the works 2-core by plurality K&K subgroup (left) and dominant structural NMF factor (right). Ribbons take the color of their origin node's group; grey marks the 16 works whose codes tie across subgroups. The single-work subgroups D-III and D-IV sit wherever their one work sits.

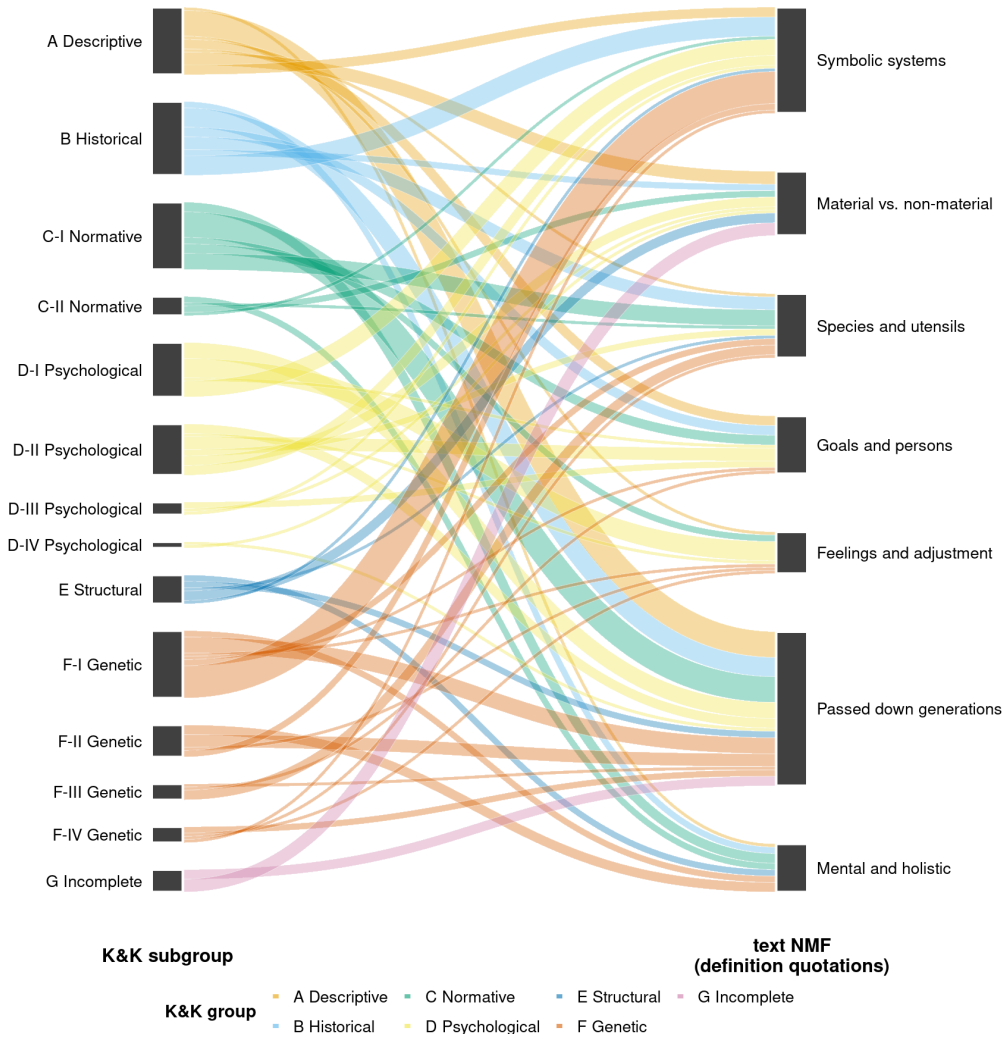


Figure 10: The 167 definition codes carrying a text, by their own K&K subgroup (left) and dominant text NMF factor (right). Ribbons take the color of their origin node's group.